

Study Questions: Systems

1. Define a Markov Blanket formally. What conditional independence relation does it encode?
2. Write the factorized joint distribution $p(\eta, s, a, \mu)$ for a system with Markov Blanket $\{s, a\}$. Explain each factor.
3. What is the Langevin equation? How does it describe the stochastic dynamics of a system?
4. How do the Markov Blanket conditions constrain the flow field $f(x)$ in the Langevin equation?
5. Define variational free energy F . Write the mathematical expression and identify each term.
6. Prove that $F = D_{\text{KL}}[q(\eta) \parallel p(\eta|s)] - \ln p(s)$. Why is this decomposition important?
7. Why is minimizing F equivalent to maximizing the evidence lower bound (ELBO)?
8. What is the recognition density $q(\eta)$? What role does it play in variational inference?
9. Why is exact Bayesian inference (computing $p(\eta|s)$ directly) typically intractable? How does variational inference address this?
10. What is the KL divergence $D_{\text{KL}}[q \parallel p]$? Prove that it is always non-negative.
11. Explain the relationship between surprise ($-\ln p(s)$) and entropy. When is surprise high?
12. How does the free energy decompose into accuracy and complexity terms? Write the expression.
13. What is a nonequilibrium steady-state density $p^*(x)$? How does it relate to self-organization?
14. How does the Fokker-Planck equation describe the evolution of the probability density over states?
15. What is the relationship between the flow field $f(x)$ and the gradient of surprise (solenoidal vs. curl-free flow)?
16. How does the Helmholtz decomposition separate the flow into dissipative and conservative components?
17. What happens to variational free energy when $q(\eta)$ exactly equals the true posterior $p(\eta|s)$?

18. Compare the free energy bound with the Helmholtz free energy in statistical thermodynamics. What is the analogy?
19. How does the choice of variational family for $q(\eta)$ (e.g., mean-field, Gaussian) affect the tightness of the free energy bound?
20. Derive the conditions under which minimizing F with respect to q recovers exact Bayesian inference.